Limit: Short Notes

How to use these notes: read the Formulas box, follow the Steps and Key points, and avoid the Watch out points.

1. Meaning of limit, LHL and RHL

Formulas

$$\text{LHL} = \lim_{h\to 0} f(a-h), \qquad \text{RHL} = \lim_{h\to 0} f(a+h)$$

KEY POINTS

- Limit exists $\iff$ LHL = RHL = finite number
- $f(a)$ (value, open/closed dot) has no role in the limit
- One side outside the domain: limit = the other one-sided limit
- From a graph: follow the curve from each side and read the height

2. Indeterminate forms

Formulas

$$\frac{0}{0},\ \frac{\infty}{\infty},\ \infty-\infty,\ 0\times\infty,\ 1^{\infty},\ 0^{0},\ \infty^{0}$$

KEY POINTS

- 0 and 1 in the forms are approached, not exact
- Exact $0\times$ anything $= 0$; (exact $1)^{\text{any}} = 1$; $(\neq 0)^{\text{exact } 0} = 1$
- $\frac{k}{\to 0}$ $(k \neq 0)$ is not indeterminate: it is $\pm\infty$ (check sides)

3. Algebra of limits

Formulas

$$\lim(f \pm g) = l \pm m, \quad \lim fg = lm, \quad \lim \frac{f}{g} = \frac{l}{m}\ (m \neq 0)$$

KEY POINTS

- Valid only when both l, m are finite
- Both limits absent: the combination may still have a limit ($[x] + \{x\} = x$)
- One exists, other not: $f \pm g$ never exists; fg, f/g may

Watch out

Split a limit into parts only if every part is finite

4. Factorisation, rationalisation, substitution

Formulas

$$a - b = \frac{a^2-b^2}{a+b} = \frac{a^3-b^3}{a^2+ab+b^2}$$

STEPS

1. Put $x = a$; if determinate, stop
2. Polynomials $\frac{0}{0}$: cancel $(x-a)$ (repeat if needed)
3. Roots: multiply by the conjugate (square or cube)
4. $\infty - \infty$: LCM or rationalise to get $\frac{0}{0}$ or $\frac{\infty}{\infty}$
5. $x \to a$: $x = a + h$; $x \to \infty$: $x = \frac{1}{t}$; $x \to -\infty$: $x = -t$

Watch out

$\sqrt{x^2} = |x| = -x$ when $x \to -\infty$

Put the value of a factor only in a product, and only if its limit is finite non-zero

5. Infinity/infinity form

Formulas

$$\lim_{x\to\infty}\frac{a_0x^m+\dots}{b_0x^n+\dots}=\frac{a_0}{b_0}\ (m=n),\ 0\ (m<n),\ \pm\infty\ (m>n)$$

KEY POINTS

- Divide by the highest power of x (or the largest base a^n)
- $\ln x \ll x^n \ll e^x$ as $x\to\infty$; $x\ln x\to 0$ as $x\to 0^+$
- Bounded $\div\infty\to 0$ (e.g. $\frac{\sin x}{x}\to 0$ as $x\to\infty$)

6. Standard limits (algebraic, trigonometric)

Formulas

$$\lim_{x\to a}\frac{x^n-a^n}{x-a}=na^{n-1},\qquad \lim_{x\to 0}\frac{(1+x)^n-1}{x}=n$$

$$\lim_{x\to 0}\frac{\sin x}{x}=\frac{\tan x}{x}=\frac{\sin^{-1}x}{x}=\frac{\tan^{-1}x}{x}=1,\qquad \lim_{x\to 0}\frac{1-\cos x}{x^2}=\frac{1}{2}$$

KEY POINTS

- $s\to 0$: $\sin s,\tan s,\sin^{-1}s,\tan^{-1}s\to s$; $1-\cos s\to\frac{s^2}{2}$; $(1+s)^n-1\to ns$
- $x^p\sin\frac{1}{x}\to 0$ $(p>0)$, does not exist $(p\le 0)$
- Angles in radians; ITF away from 0: put ITF $=\theta$

Watch out

Replace only factors, never terms of a sum or difference

7. Exponential and logarithmic limits

Formulas

$$\lim_{x\to 0}\frac{e^x-1}{x}=1,\quad \lim_{x\to 0}\frac{a^x-1}{x}=\ln a,\quad \lim_{x\to 0}\frac{\ln(1+x)}{x}=1$$

KEY POINTS

- $s\to 0$: $e^s-1\to s$, $a^s-1\to s\ln a$, $\ln(1+s)\to s$
- $S\to 1$: $\ln S\to S-1$, $S^n-1\to n(S-1)$
- $a^x-b^x=(a^x-1)-(b^x-1)$ before replacing

8. One raise to infinity

Formulas

$$\lim f^g=e^{\lim g\,(f-1)}\quad (f\to 1,\ g\to\infty)$$

KEY POINTS

- Check: base $\to 1$ (not exact 1), exponent $\to\infty$
- 0^0, ∞^0 and general f^g: $e^{\lim g\ln f}$

9. Expansions

Formulas

$$\sin x=x-\frac{x^3}{6}+\frac{x^5}{120},\ \cos x=1-\frac{x^2}{2}+\frac{x^4}{24},\ \tan x=x+\frac{x^3}{3}+\frac{2x^5}{15}$$

$$e^x=1+x+\frac{x^2}{2}+\frac{x^3}{6},\ \ln(1+x)=x-\frac{x^2}{2}+\frac{x^3}{3},\ \tan^{-1}x=x-\frac{x^3}{3}+\frac{x^5}{5}$$

$$(1+x)^n=1+nx+\frac{n(n-1)}{2}x^2,\quad \sin^{-1}x=x+\frac{x^3}{6}+\frac{3x^5}{40}$$

KEY POINTS

- Four standard: $\frac{x-\sin x}{x^3} \to \frac{1}{6}$, $\frac{\tan x - x}{x^3} \to \frac{1}{3}$, $\frac{e^x-1-x}{x^2} \to \frac{1}{2}$, $\frac{\ln(1+x)-x}{x^2} \to -\frac{1}{2}$
- Lowest powers: equal $\Rightarrow$ ratio of coefficients; numerator higher $\Rightarrow 0$; denominator higher $\Rightarrow$ does not exist
- Make the variable $\to 0$ before expanding

10. Finding unknown constants

STEPS

1. Expand numerator (and denominator) up to the needed power
2. Finite limit: coefficients of powers below the denominator's lowest power $= 0$
3. Matching power: coefficient ratio = given value
4. Without expansion: limit finite and denominator $\to 0 \Rightarrow$ numerator $\to 0$

11. Limits that are derivatives

Formulas

$$f'(a) = \lim_{h\to 0} \frac{f(a+h)-f(a)}{h}, \quad f''(x) = \lim_{h\to 0} \frac{f(x+2h)-2f(x+h)+f(x)}{h^2}$$

KEY POINTS

- Higher orders: coefficients of $(h-1)^n$
- Given $f(a)$, $f'(a)$: add and subtract $f(a)$ to create $\frac{f(x)-f(a)}{x-a}$

12. Modulus, [ ], , sgn

Formulas

$$x \to n^-: \ [x] = n-1, \ \{x\} \to 1; \qquad x \to n^+: \ [x] = n, \ \{x\} \to 0$$

KEY POINTS

- Always find LHL and RHL separately
- Inside $\to$ non-integer: just substitute
- Inside $\to$ integer: decide whether it comes from below or above
- $\frac{\sin x}{x}, \frac{\tan^{-1} x}{x}, \cos x \to 1^-$; $\frac{\tan x}{x}, \frac{\sin^{-1} x}{x} \to 1^+$
- Rational/irrational definitions: limit only where both definitions agree

13. Sandwich theorem

Formulas

$$g \le f \le h, \ \lim g = \lim h = l \ \Rightarrow \ \lim f = l$$

$$x - 1 < [x] \le x$$

KEY POINTS

- Series of n terms: $n\cdot$(smallest) $\le S_n \le n\cdot$(largest)
- $\sum_{k=1}^{n} [kx]$: use $\sum k = \frac{n(n+1)}{2}$ on both bounds

Watch out

Different limits of the two bounds: theorem gives nothing

14. Composite functions and built-in limits

KEY POINTS

- $\lim f(g(x))$: find g's limit and side, then f at that value from that side
- f continuous at $\lim g$: $\lim f(g) = f(\lim g)$ (ln, e, $\sqrt{}$, sin, ITF)
- $\lim_{n\to\infty} x^n = 0$ $(|x|<1)$, 1 $(x=1)$, ∞ $(x>1)$, does not exist $(x \le -1)$
- Built-in limit: split into $|x|<1$, $|x|=1$, $|x|>1$ and divide by the dominating term

15. Geometrical limits

KEY POINTS

- Write lengths/areas in the angle: sector $\frac{1}{2}r^2\theta$, triangle $\frac{1}{2}ab\sin C$, tangent length $r\tan\frac{\theta}{2}$
- Then use standard limits or expansions ($\theta - \sin\theta \approx \frac{\theta^3}{6}$)

16. Common mistakes checklist

STEPS

1. Did I check the form before simplifying?
2. Did I replace $\sin s \to s$ etc. only for factors, not inside sums?
3. Did I find LHL and RHL for [ ], { }, | |, sgn, ITF, $e^{1/x}$?
4. Did I use $\sqrt{x^2} = -x$ for $x \to -\infty$?
5. Did I expand far enough (up to the denominator's power)?
6. Is the base of 1^∞ really approaching 1 (not exactly 1)?
7. Is my answer finite? An infinite value means the limit does not exist